Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			weigh **6.21 g** of sodium thiosulfate (into a beaker) (1) credit each of the following points • dissolve (in deionised water) • transfer • 250 cm^3 volumetric flask • rinse beaker / funnel / stirring rod (with deionised water) • make up to the mark (with deionised water) • invert several times to mix solution well award (3) for all **six** points award (2) for any **four** points award (1) for any **two** points	3	1		4	1	4
	(b)	(i)		award (1) for either of following sulfur forms as solid precipitate of sulfur forms			1	1		1
		(ii)		time increased as less depth of solution (less sulfur formed over cross) OWTTE			1	1		1
	(c)	(i)		award (1) for range of volumes of thiosulfate – all lower than 40 cm^3 award (1) for volumes of water to give total volume 40 cm^3 (thiosulfate + water) **and** 10cm^3 volume of HCl (1)		2		2		2
		(ii)		s^{-1} ignore 1000		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		rate is directly proportional to volume of thiosulfate		1		1	1	
	(d)	(i)		rate at 30°C is 12 (1) time = $\frac{1000}{12}$ = 83 s (1) ecf possible	2			2	2	
		(ii)		this is true only to some extent here rate doubles when temperature goes from 20°C to 30°C (1) [also when temperature goes from 10°C to 20°C] rate <u>more than</u> doubles when temperature goes from 30°C to 40°C (1)			2	2	1	
		(iii)		particles have more energy (1) neutral answer – particles move faster more collisions exceed activation energy (1) neutral answer – more successful collisions	2			2		
				Question 10 total	**7**	**5**	**4**	**16**	**5**	**8**